MIL-W-52187C
28 May 1975
SUPERSEDING
MIL-W-52187B
30 June 1969

MILITARY SPECIFICATION

WATER PURIFICATION PLANT, SEMIPERMANENT

ELECTRIC-MOTOR-DRIVEN, ERDLATOR-TYPE

CLARIFIER, SAND FILTERS, 10,000-GPH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a military design electric-motor-driven water purification plant, with erdlator-type clarifier and sand filters, capable of furnishing potable water at a rate of 10,000 gallons per hour.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial).
NN-P-530	- Plywood, Flat Panel.

FSC 4610

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

TT-E-522	- Enamel, Phenolic, Lusterless, Outside.
TT-P-664	- Primer; Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-35	- Bags, Textile-Shipping, Burlap Cotton and Waterproof Laminated.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

Military

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2638	- Lacquer, Vinyl Resin, Gasoline and Water Resistant.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-W-6858	- Welding, Resistance: Aluminum, Magnesium, Nonhardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys; Spot and Seam.
MIL-T-10086	- Tanks, Liquid Storage, Metal; Vertical, Bolted.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-P-15328	- Primer (Wash), Pretreatment, Blue, (Formula No. 117-BF or Metals).
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

MIL-L-21260	- Lubricating Oil, Internal - Combustion Engine, Preservative and Break-In.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-P- 52405	- Pumps, Centrifugal, Electric-Motor-Driven (For Water).

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-147	- Palletized and Containerized Unit Loads 40" X 48" Pallets, Skids, Runners, or Pallet-Type Base.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MS35844	- Plug, Machine Thread; Magnetic: Drain.
MS49006	- Plugs, Pipe, Magnetic-Headless; Aluminum, Iron or Steel.

DRAWINGS

ME

TA13220E8868	- Water Purification Plant, Semi-Permanent, 10,000 GPH.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

COUNTY OF LOS ANGELES - AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents.

(Application for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B 16.1 - Cast Iron Pipe Flanges and Flanged Fittings.
B 16.21 - Nonmetallic Gaskets for Pipe Flanges.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

AMERICAN WELDING SOCIETY, INC. (AWS)

A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.9 - Corrosion-Resisting Chromium and Chromium-Nickel Steel Welding Rods and Bare Electrodes.
A5.10 - Aluminum and Aluminum-Alloy Welding Rods and Bare Electrodes.
D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society Inc., 2501 NW Seventh Street, Miami, FL 33125.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION

MG-1 - Motor and Generators.

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 East 44th Street, New York, NY 10017.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15086.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago IL, 60606.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description. The water purification plant (hereinafter called "plant") shall be in accordance with the Top Assembly TA13220E8868 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish a plant for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce plants that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the supplier shall furnish to the Government a plant for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate departmental medical agency who will act as an advisor to the procuring agency (see 6.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

3.5 Maintainability All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage. The plant shall exhibit no degratation in performance after maintenance operations.

3.6 Safety. Unless otherwise specified on the applicable drawings, nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided.

3.7 Float valves. The float valves, as specified on applicable drawings, shall be of the single-seated, chatter-resistant type with semisteel body and brass internal working parts. The valve shall be designed for working pressures up to 100 pounds per square inch (psi) and shall be suitable for cold-water service. The valve shall have an internal piston disk holder, pilot valve, a renewable disk of composition rubber, a screwed-in renewable seat of a diameter equal to the nominal pipe size of the valve, and a swivel yoke. The inlet and outlet connections shall be flanged. Functional test shall be as specified in 4.6.2.10.2.

3.8 Erdlator agitator assembly.

3.8.1 Adjustable motor base. The base, as specified on the applicable drawing, shall permit 4.31 inches horizontal adjustment normal to the motor shaft and 1.25 inches horizontal adjustment parallel to the motor shaft. The 4.31-inch adjustment shall be by handwheel operation through an extended shaft integral with the motor base. The 1.25-inch adjustment shall be by slotted holes for mounting the motor. Functional test shall be as specified in 4.6.2.1.

3.8.2 Reduction gear assembly. The reduction gear assembly, as specified on the applicable drawing, shall be of the worm-gear, reduction type with a vertical, low-speed shaft extended downward. The gear assembly shall operate at a speed variation from not less than 50 nor more than 180 rpm at the output shaft.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

The horsepower rating shall be not less than 2.75 for a Class II service classification at a worm shaft speed of 1,750 rpm and an output shaft speed of 180 rpm. The input shaft shall be not less than 1.125 inches long. The output shaft shall extend not more than 2.25 inches below the base. Functional test shall be as specified in 4.6.2.1.

3.8.3 Variable-speed pulley. The variable-speed pulley, as specified on the applicable drawing, shall be of the expandable type and shall provide a variation of speed of from not less than 50 rpm nor more than 180 rpm at the output shaft of the reduction gear assembly for the agitator. Functional test shall be as specified in 4.6.2.1.

3.9 Raw-water-flow indicatior. The raw-water-flow indicator shall be of the orifice or the in-line, variable-area type and shall indicate flow rates of from 0 to 12,000 gph with an accuracy of plus or minus 3 percent of the actual flow rate in the range of 6,000 gph to 12,000 gph, when tested as specified in 4.6.2.2.

3.9.1 Orifice type.

3.9.1.1 Indicating mechanism. The indicating mechanism shall indicate flow rates from 0 to 12,000 gph. The mechanism shall operate by use of rupture-proof bellows and shall incorporate a torque tube drive which shall not require packing or lubrication maintenance. The mechanism shall be responsive to any change in differential pressure, shall have positive protection against over-ranging, and shall not be affected by changes in atomspheric temperatures. The mechanism shall be easy to read and shall be adjustable to zero reading without removal of the cover plate or disturbance of the pointer. Vent cocks and drain cocks shall be provided for the complete removal of entrapped air during filling and the complete drainage of the mechanism without removal of the mechanism from its mounting.

3.9.1.2 Orifice flanges. The orifice flanges shall be iron castings, shall have a pressure rating of 125 psi, 2.0-inch threaded, 11.50 NPT, and shall clamp and center the orifice plate (see 3.9.1.3) between the faces. The pressure-taps shall be corner taps and shall be drilled and tapped 0.25-inch NPT.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

3.9.1.3 Orifice plate. The orifice plate shall be of AISI Type 316 corrosion-resisting steel and shall be suitable for measuring flow rates from 0 to 12,000 gph. The orifice plate shall be provided with an identification plate stamped with the information pertinent to the particular installation. The identification plate shall be riveted to the orifice plate and shall be visible after the orifice plate is installed between the orifice flanges.

3.9.2 In-line, variable-area type. The in-line, variable-area type shall consist of a variable-area metering tube, metering float, indicating mechanism, and two detachable extensions. The variable-area metering tube shall be of cast aluminum, and shall be complete with removable combination float guides and stops of AISI Type 316 corrosion-resisting steel. The metering float shall have top and bottom extensions and shall be of AISI Type 316 corrosion-resisting steel. The indicating mechanism shall be magnetically coupled to the metering float without the use of stuffing boxes or torque tubes. The detachable extensions shall be cast aluminum.

3.10 Filter sand agitator assembly. The filter sand agitator, as specified on the applicable drawing, shall be a water-propelled, revolving type for a circular gravity sand filter, consisting of a central bearing of bronze construction, and a bronze tie with arms extending laterally, giving an overall length of 60-inches and containing eight nozzles spaced to effectively cover the surface area of the filter sand when the agitator is in operation. Each arm shall be equipped with end nozzles, one pointing out and down at an angle of 15 degrees with the horizontal to direct a stream of water to the outer edge of the filter and one pointing in toward the center and parallel to the arm. Arms shall be red brass; nozzles, bearings, and other fittings shall be bronze. Nozzle size shall be adequate to give a water stream of not less than 0.50 gpm per sq. ft. of filter area at a pressure of not less than 50 psi. The bearing supply connection shall be internal 1.50-inch NPT. Functional test shall be as specified in 4.6.2.3.

3.11 Filter sand and gravel. Filter sand and gravel shall be as specified on the applicable drawing.

3.12 Chemical solution tanks. When fabricated as specified on the applicable drawings, the chemical solution tanks shall be tested as specified in 4.6.2.6.

3.13 Filter tanks. When fabricated as specified on the applicable drawings, the filter tanks shall be tested as specified in 4.6.2.3.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:07Z
Check the source to verify that this is the current version before use.

3.14 Concentrator tank assembly. When fabricated as specified on the applicable drawing, the concentrator tank assembly shall be tested as specified in 4.6.2.5.

3.15 Wetwell tank assembly. When fabricated as specified on the applicable drawing, the wetwell tank assembly shall be tested as specified in 4.6.2.6.

3.16 Loss-of-head indicators. When fabricated as specified on the applicable drawing, the loss of head indicators shall be tested as specified in 4.6.2.7.

3.17 Aeration tank assembly. When fabricated as specified on the applicable drawing, the aeration tank assembly shall be tested as specified in 4.6.2.8.

3.18 Limestone slurry feeder. When fabricated as specified on the applicable drawing, the limestone slurry feeder shall be tested as specified in 4.6.2.9 and on the applicable drawing.

3.19 Pumps. Pumps and pump motors shall conform to MIL-P-52405, classification as follows:

(a) Wash water - Type II, Class C
(b) Filter agitator - Type II, Class B
(c) Raw Water - Type I, Class A
(d) Distribution - Type I, Class A

Pumps shall conform dimensionally to the applicable drawing, or the supplier shall furnish all pipe, fittings, and other items that will be required to place the pumps in the system as shown on **equipment and installation drawings. Pumps shall be furnished with anchor bolts, washers, and nuts of the size and quantity required to fasten the pumps to the concrete pads as shown on the installation drawings.**

3.20 Float-actuated switches. The float-actuated switches for mounting on the raw-water and distribution storage tanks specified on the applicable drawings shall operate on 115 volts, 15 amperes, 60 hertz. The switch mechanism shall be of the snap-acting, magnetically-operated type and shall be inclosed in a weather-resisting housing. The switch for the raw-water tank shall be normally open at high water level. The switch for the distribution tank shall be normally closed at high water level. Each switch shall be supplied complete with rod and float, for mounting and operating on a bolted, steel tank conforming to MIL-T-10086, 10,500-gallon size, (250 bbl).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

3.21 Agitator motor. The electric motor for the erdlator agitator shall conform to NEMA Standard MG 1, Code B, Design A, continuous-duty rating at 105° C, waterproof enclosure, Frame 184T, Class H insulation, 1,800 rpm synchronous, dual voltage 208/416 vac, 60 hertz, 3 phase, maximum full load current 7.1/2.9 amperes, output 1.5 hp. Motor bearings shall be of the ball or roller-type, designed to carry all radial and thrust loads, and shall be installed in a permanently lubricated housing rated for continuous operation condition. The motor shall be provided with either a watertight terminal connection compartment integral with the motor frame, or a watertight terminal box complete with a watertight stuffing tube connector.

3.22 Wiring diagram and calibration charts. A wiring diagram and calibration charts shall be furnished with each plant. The diagram and charts shall have a hole at each corner for use with a No. 10 slotted, roundhead wood screw. Four No. 10 cadmium plated wood screws 0.50-inch long shall be furnished with each diagram and chart. The calibration flow chart shall be as specified on the applicable drawing.

3.22.1 Wiring diagram. The wiring diagram shall be not less than 24 inches by 24 inches and shall conform to TA13220E8868. The wiring diagram shall reflect the wiring hookup actually required.

3.23 Limestone slurry feeder chart. A limestone slurry feeder flow chart, covering the full range for a raw-water-flow rate of 10,000 gph, shall be furnished as specified on the applicable drawing.

3.24 Accessories. The following additional accessories shall be furnished with each plant:

Item	Quantity
(a) Overflow pipe, conforming to Part Number 11004-55-5.	2
(b) Flange, reducing, ANS-B16.1, 4 X 10.50, cast iron.	4
(c) Gasket, for ANS-B16.21, 4 X 10.50, reducing flange, cloth inserted rubber.	4
(d) Flange, for float valve support, Part Number 11004-26-1.	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

3.25 Electromagnetic interference. The electromagnetic interference emission and susceptability characteristics of the plant shall conform to MIL-STD-461, Class II A.

3.26 Lubrication. All surfaces not in contact with the water being treated and requiring lubrication shall be provided with a means for lubricating.

3.26.1 Lubricants. The plant shall operate as specified herein when lubricated with military lubricants (see 6.9).

3.26.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, Types I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.26.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.26.4 Filling, draining, and checking provisions. Enclosures such as gear cases that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 0.50-inch pipe size, or sight gages to determine the level of lubricant. Each inclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a magnetic drainplug conforming to MS35844 or MS49006. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Integral tubes or troughs shall be used to convey the lubricant from the drain to a point where it can be collected in a container. The drain plug, the filling means, and the lubricant level dipsticks or holes shall be accessible without disturbance of accessories or parts.

3.26.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with grease as follows:

(a) MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250° F.
(b) MIL-G-10924 - For other bearings where operating temperatures do not exceed 175° F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

The plant components shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication because military greases are not compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.26.6 Oil lubrication. Unless otherwise preserved for packaging (see 5.1), and upon completion of all testing, all components shall be filled to the operating level with winter grades of military oils designated for use in the temperature range of 0° F to plus 32° F. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.27 Fungus and moisture resistance. The electrical circuity, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.27.1, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.27.1 Composition. Composition I shall be used unless Composition II must be used in order to comply with local air pollution regulations governing the application of the varnish. When Composition II is used, the supplier shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66 - Air Pollution Control District, County of Los Angeles, CA.

3.28 Identification marking. The plant shall be identified in accordance with MIL-STD-130. The marking shall be applied to the plant on plates conforming to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 material. The plate shall be attached to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

plant or major components with adhesive specified on applicable drawing or if appropriate be attached by screws, bolts, or rivets in a conspicuous protected location.

3.29 Shipping data plate. Shipping data plate shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material. The plate shall show the silhouette of the plant or major components in transport position indicating the location and capacity of the lifting and tiedown points. The plate shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.30 Instruction plates. Each plant shall be provided with instruction plates to include but not limited to warning, cautions, and safety hazards. Also to describe any special or important procedures to be followed in assembling, operating, or servicing of the plant. Plates shall conform to MIL-P-514, Type III, Composition C of Type I, Grade A, Class 1 material, size as applicable, and shall have corner holes suitable for attaching or hanging.

3.31 Treatment and painting. Unless otherwise specified on the applicable drawings, all ferrous metal parts not in contact with potable water, chemical or slurry solutions, shall be cleaned, treated and painted in accordance with MIL-T-704, Type B, except that the finish coat shall conform to TT-E-522, Color No. 36231 of FED. STD. No. 595. The direction of rotation of each pump shall be permanently stenciled on top of each pump casing with a contrasting color. Finished colors for piping shall be as specified on the applicable drawing.

3.32 Government-furnished property. **When specified (see 6.2), ancillary items as required, referenced in the appendix, will be furnished by the Government for each plant.**

3.33 Performance. The plant shall have the following performance characteristics:

(a) The plant shall be capable of operation in accordance with the preproduction requirements and when tested as specified herein.
(b) The plant components shall meet performance requirements as specified herein and on the applicable drawings.

The performance tests shall be as specified in 4.6.2.10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

3.34 Workmanship. All parts, components, and assemblies of the plant including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All edges shall be rounded or chamfered. The plant shall be free from such defects as misalined components, incomplete welds, rust, extraneous material, cracks, leaks, and other defects that could impair the operation and serviceability of the plant.

3.34.1 Metal fabrication. Metal used in the fabrication of the plants shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Sheared, sawed, and chipped edges shall be free of burrs and slivers. Flame cutting, using a tip suitable for the thickness of the metal, may be employed instead of shearing or sawing for steel only. Methods utilizing heat shall not be used for nonferrous metals. Burned surfaces of flame-cut meterial shall be ground or machined to remove ash and cooling checks. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.34.2 Bolted connections. Unless otherwise specified on the applicable drawing, boltholes shall be accurately formed and shall have the burrs removed. Washers or lock-washers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J 429 or cap screws shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.34.3 Riveted connections. Unless otherwise specified on the applicable drawing, rivets shall fill the holes completely. All rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member and shall be in accordance with SAE J 429.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

3.34.4 Welders, weldments, and welding.

3.34.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular codes:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Suppliers who only make horizontal welds need not qualify welders for **"all position welding". Subject to approval by the Government, the** supplier's standard welder qualification may be substituted in lieu of the above codes provided that the supplier's procedure is equivalent to the above codes. The supplier shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds.

3.34.4.2 Weld filler metals. Unless otherwise specified on the applicable drawing, aluminum filler metal shall be in accordance with AWS A5.10, Type ER5356 or ER5556, except that filler metal ER4043 may be used in highly restricted locations if conditions cannot be eliminated by adjustments of procedures of fixtures, or when joining base metal Type 6061 to other types of aluminum. Steel filler metal shall be in accordance with AWS A5.1, Type E60610 or E6011 for carbon steel; Type E7018 for high-strength, low-alloy steel; Type E6018 for high-strength, low-alloy steel to carbon steel; and AWS A5.9, Type ER308L for corrosion-resisting steel.

3.34.4.3 Aluminum and steel weldments. The surfaces and edges of parts to be welded including layers of weld materials when multiple layers are required shall be free from oxide scale, paint, lubricant, and other foreign matter. Welds shall transmit stress to the heat-affected zone without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. All welded parts shall be free from cracks, porosities and other imperfections that may reduce the effectiveness of the part. Welds involving corrosion-resistant steel shall be passivated and annealed as specified on the applicable drawings.

3.34.4.3.1 Aluminum welding. Unless otherwise specified on the applicable drawing, aluminum welding, except spot and seam welding, shall be done by the metal-inert-gas (MIG) method or pulse-arc methods that will provide equivalent mechanical properties of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

filler metal to base metal combinations as specified in 3.34.4.2. All welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metals shall be obtained on all welds without metallic or nonmetallic inclusions. Finished aluminum welds may be ground, filed, chipped, or wire-brushed, but shall not be hammered. Preheating for welding is permissible provided the temperature does not exceed 300° F for a total time not more than 30-minutes. No process requiring the use of welding flux shall be used. Work shall be positioned and jigged for flat welding whenever practicable.

3.34.4.3.2 Spot and seam welding. Spot and seam welding shall be in accordance with MIL-W-6858, Class C.

3.34.4.3.3 Steel welding. Unless otherwise specified on the applicable drawing, steel welding shall be done by the shielded-electric-arc method or metal-inert-gas (MIG) process in accordance with qualification procedures prescribed in 3.34.4.1, or by any other method providing equivalent mechanical properties of the filler metal to base metal combinations as specified in 3.34.4.2. Where embrittlement of the joined metal may result, preheating or annealing or both shall be employed. Work shall be positioned for flat welding whenever practicable. Complete and uniform penetration and fusion of the metals shall be obtained on all welds.

3.34.4.4 Weldment specimens. The supplier shall submit steel and aluminum specimens representing each typical weld joint utilized in fabrication of the plant. These specimens shall be furnished at the same time as the preproduction model plant. Weldment specimens shall be submitted by each welder performing weldments on the plant with each welding equipment used. Acceptable specimens shall represent the minimum acceptable weld quality and shall be retained as an inspection standard for evaluation of production weldments.

3.34.4.5 Welding procedure. Prior to any welding, the supplier shall prepare and submit to the contracting officer for approval a written welding procedure defining process and control. This procedure shall include, but not be limited to, the factors listed in Table I. Welding procedure changes affecting any of the factors listed in Table I shall be submitted to the contracting officer for approval prior to adoption of the change.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

Table I. Factors to be Included in Joint Welding Procedure

Item No.	Factor
1.	Welding process and techniques including shielding medium.
2.	Welding process operation.
3.	Type of current, ac or dc; and polarity of dc, if used.
4.	Shielding gas composition.
5.	Joint design, as specified.
6.	Method of joint preparation.
7.	Preweld and interpass cleaning.
8.	Preheat or ambient temperature range.
9.	Composition of filler metal, as specified.
10.	Filler metal size for all passes.
11.	Number and sequence of passes.
12.	Welding position, to be flat whenever practicable.
13.	Repair method to be used in removing defects.
14.	Method of inspection used to insure proper repair.
15.	Method of qualification of welders.
16.	Method of determining capability of automatic equipment and equipment operator.
17.	Workmanship specimens.
18.	Method of inspection of finished welds.

3.34.4.6 Rewelding defects. Portions of welded joints that have been rejected on inspection because of defects specified in 3.34.4.3 or not conforming to the weldment specimens specified in 3.34.4.4 may be repaired only by rewelding. The defective weld shall be removed by chipping or machining. With the exception of steel, flame cutting shall not be used. Before rewelding, the joint shall be inspected to assure that all the defective weld has been removed and that the joint is accessible so that the welding operator can obtain full preparation and fusion through the joint. Repairs in the form of patching aluminum shall not be performed.

3.34.4.7 Jigs and fixtures. Components except minor parts shall be assembled in jigs or frames and welded while held in proper position. Jigs or frames shall be designed to minimize distortion of the parts being welded.

3.34.5 Diagrams, charts, and plates. Diagrams, charts, and plates attached to components with adhesive shall be applied to give a surface free of wrinkles, bubbles, or other defects that may cause the diagrams, charts, and plates to become loose or damaged.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:07Z
Check the source to verify that this is the current version before use.

3.34.6 Threads. Unless otherwise specified on the applicable drawing, threads shall be in accordance with Handbook H28.

3.34.7 Cleaning. The plant shall be thoroughly cleaned of all cutting oil, residue of tests, grease, dirt, metal scrap, and other contaminants. Cleaning shall be accomplished in a manner that will not leave a residue or otherwise make the plant unsuitable for its intended use. The plant shall be completely drained before preparation for delivery.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection. Failure of any test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.1.3 Parts and components. Parts and components shall be in accordance with the Supplementary Quality Assurance Provisions (SQAP's) shown on the drawings. The drawings specify the characteristics requiring inspection, the sampling plan, and the basis for acceptance and rejection (see 6.8).

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial product inspection (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:07Z
Check the source to verify that this is the currentversion before use.

(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery(see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The plant shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The plant shall be tested as specified in 4.6.2.1 through 4.6.2.13. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), an initial production plant will be selected at random by the Government from the plants being produced from production tooling and shall be examined as specified in 4.6.1 and tested as specified in 4.6.2.1 through 4.6.2.13 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production plant shall not exclude the remaining plants from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test. the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production plant to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for refusal by the Government to continue acceptance of production plants until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on plants previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all plants accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each plant shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

3.34.6 Threads. Unless otherwise specified on the applicable drawing, threads shall be in accordance with Handbook H28.

3.34.7 Cleaning. The plant shall be thoroughly cleaned of all cutting oil, residue of tests, grease, dirt, metal scrap, and other contaminants. Cleaning shall be accomplished in a manner that will not leave a residue or otherwise make the plant unsuitable for its intended use. The plant shall be completely drained before prepara for delivery.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Ex as otherwise specified in the contract or order, the supplier may u his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of th inspections set forth in the specification where such inspections a deemed necessary to assure supplies and services conform to prescri requirements.

4.1.1 Component and material inspection. The supplier is resp sible for insuring that components and materials used are manufactu examined, and tested in accordance with referenced specifications a standards.

4.1.2 Disassembly inspection. Failure of any test by the preproduction model shall be cause for disassembly, in the presence a Government representative, of the preproduction model to the exte necessary to determine the cause of the failure. Each disassemble shall be examined in detail for compliance with this specification referenced drawings in regard to materials, dimensions, tolerances workmanship. Parts not complying with such requirements shall be rejected.

4.1.3 Parts and components. Parts and components shall be in accordance with the Supplementary Quality Assurance Provisions (SQ shown on the drawings. The drawings specify the characteristics r ing inspection, the sampling plan, and the basis for acceptance an rejection (see 6.8).

4.2 Classification of inspections. Inspections shall be clas as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial product inspection (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:07Z
Check the source to verify that this is the current version before use.

(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery(see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The plant shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The plant shall be tested as specified in 4.6.2.1 through 4.6.2.13. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), an initial production plant will be selected at random by the Government from the plants being produced from production tooling and shall be examined as specified in 4.6.1 and tested as specified in 4.6.2.1 through 4.6.2.13 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production plant shall not exclude the remaining plants from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test. the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production plant to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for refusal by the Government to continue acceptance of production plants until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on plants previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all plants accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each plant shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

4.5.2 Tests. Each plant shall be tested as specified in 4.6.2.1 through 4.6.2.13. Failure of any test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The plant shall be examined for the following defects:

Major

101. Dimensions not as specified.
102. Fasteners inadequate.
103. Assembly incomplete or incorrect.
104. Parts or components missing.
105. Parts or components do not function properly.
106. Material not as specified.

Minor

201. Poor fit at joints, seams, edges; misalinement.
202. Lubrication devices not accessible.
203. Dirty components.
204. Poor treatment and painting (incomplete preparation before painting; scale, lumps, runs, craze, thin spots, bare spots; color of dry film not as specified; or streaked and uneven color).
205. Accessories, identification or other marking, operating and safety instructions, or technical publications missing.
206. Identification or other marking misleading.

4.6.2 Tests. Components of the plant shall be tested as follows.

4.6.2.1 Erdlator and agitator assembly. The erdlator shall be assembled, and all side and bottom openings shall be plugged leaktight. The agitator motor shall be connected to an appropriate power source. The erdlator assembly shall be filled approximately one-third full with tap water, the agitator motor and gear reducer assembly shall be started, and the rpm of the motor and the output shaft of the gear reducer shall be measured with a tachometer at various pulley settings to determine conformance to 3.8 through 3.8.3. The agitator motor shall be stopped and the erdlator assembly completely filled with water. The assembly shall remain completely filled for not less than 1 hour and then examined for leaks. Evidence of leakage or nonconformance to 3.8 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

4.6.2.2 Raw-water-flow indicator. The raw-water-flow indicator shall be tested at flows of 6,000, 8,000, 10,000 and 12,000 gph, by collecting the volume of water flowing through the flow indicator for 1-minute. The arithmetic mean of three determinations for each of the flows to be tested shall be used for this test. Pipe used for testing shall be the same type, size and straight length as specified for the plant. Nonconformance to 3.9 shall constitute failure of this test.

4.6.2.3 Filter and sand agitator assembly. All openings shall be plugged leaktight and the assembly completely filled with tap water. During filling, the filter float shall be examined for proper operation. The assembly shall be allowed to stand completely filled for not less than 1-hour and examined for leaks. At the end of the leak examination, the assembly shall be drained until the level of the water is even with the top of the trough. The filter sand agitator shall then be connected to a water source equivalent to the filter sand agitator pump and the agitator assembly examined for proper operation. During draining, the filter float assembly shall be examined for proper operation. Filtering material need not be used for this test. Evidence of leakage or nonconformance to 3.10 and 3.13 shall constitute failure of this test.

4.6.2.4 Chemical solution tank. The chemical solution tanks shall be tested for pinholes with a wire brush attached to a transformer delivering not less than 15,000 volts. Evidence of pinholes or nonconformance to 3.12 shall constitute failure of this test.

4.6.2.5 Concentrator tank assembly. All openings shall be plugged leaktight. The assembly shall be completely filled with tap water. The assembly shall be allowed to stand completely filled for not less than 1 hour and then examined for leaks. Evidence of leakage or nonconformance to 3.14 shall constitute failure of this test.

4.6.2.6 Wetwell. All openings shall be plugged leaktight. The wetwell shall be completely filled with tap water. The wetwell shall remain completely filled for not less than 1 hour and examined for leaks. Evidence of leakage or nonconformance to 3.15 shall constitute failure of this test. The wetwell shall be assembled to the erdlator for this test.

4.6.2.7 Loss-of-head indicator. All openings shall be plugged leaktight. The assembly shall be completely filled with tap water and examined for proper operation. The assembly shall be allowed to stand completely filled for not less than 1-hour and examined for leaks. During draining, the assembly shall be examined for proper operation. Improper operation, leakage, or nonconformance to 3.16 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:07Z
Check the source to verify that this is the currentversion before use.

4.6.2.8 Aeration tank assembly. All openings shall be plugged leaktight. The assembly shall be completely filled with tap water. The assembly shall be allowed to stand completely filled for not less than 1-hour and examined for leaks. Evidence of leakage or nonconformance to 3.17 shall constitute failure of this test. The raw water trough shall be assembled to the aeration tank assembly for this test.

4.6.2.9 Limestone slurry feeder. The discharge flow rate of the limestone slurry feeder shall be tested as specified on the applicable drawing. Nonconformance to the requirements specified on the applicable drawing and 3.18 shall constitute failure of this test.

4.6.2.10 Performance (first article).

4.6.2.10.1 Test equipment. Test equipment shall include but shall not be limited to the following:

(a)	Power source	- 120/208/416, 3 phase, 4 wire, 50/60 hertz.
(b)	Centrifugal pumps	- As specified herein and on the applicable drawings.
(c)	Piping or hose	- As required to fulfill the essential tests specified.
(d)	Water	- Tap (potable from an approved source).

4.6.2.10.2 Test procedure. Test procedure shall be as specified herein. Short shutdown periods for inspection, lubrication, or functional changes shall be permitted providing normal operation is resumed within 30-minutes. For the purpose of this test procedure, a test shall be as specified in 4.6.2.1 through 4.6.2.12, for periods not less than those specified.

4.6.2.10.3 Failure criteria.

(a) Failure of any component to commence operation or degradation of performance capability below designated levels specified herein.
(b) Leaks at any welded seam, joint or any connection to cause cessation of operation.
(c) Lights, buzzers, or any electrical switching gear fails to operate.
(d) Valves fail to operate properly.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:07Z
Check the source to verify that this is the current version before use.

(e) Chemical solution feeder fails to operate as specified.
(f) Limestone slurry feeder fails to operate as specified.
(g) Serious personnel safety hazards.

Note: All components of the plant shall be tested as specified herein, and to the fullest extent practicable without assembling the 4-inch and 6-inch piping. Nonconformance to 3.33 shall constitute failure of this test.

4.6.2.11 Maintainability. Maintenance operations required during preproduction or initial production testing shall be accomplished in accordance with 3.5. Inability of the plant to be maintained utilizing the support elements shall constitute failure of this test and shall require revision of the document(s). Nonconformance to 3.5 shall constitute failure of this test.

4.6.2.12 Safety. Safety evaluation shall be accomplished throughout testing. Nonconformance to 3.6 shall constitute failure of this test.

4.6.2.13 Electromagnetic interference. The preproduction model shall be tested to determine conformance with 3.25. The supplier shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.7 Inspection comparison. The Government may select a plant at any time during the contract production period and subject this plant to the examination specified in 4.6.1 and the tests specified in 4.6.2.1 through 4.6.2.12 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial plant. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of any inspection comparison plant to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

the inspection comparison plant and shall be cause for refusal by the Government to continue acceptance of production plant until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on plants previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all plants accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.8 Inspection of preparation for delivery.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. Examine the preproduction pack for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.1.2 Test. The preproduction pack of crated components for Level A shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of not less than 10 miles nor more than 11 miles per hour. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, tie, stays, blocking, or bracing, or any visual damage to the components shall constitute failure of the preproduction pack and shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

207.	Materials, methods, and containers not as specified for Level A or B.	Each incorrect material, method, or container shall be considered one defect.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:07Z
Check the source to verify that this is the currentversion before use.

208. Preservatives applied to surfaces which will come in contact with chemical solutions, slurries, or water, except waste water.
209. Disassembly not as specified for Level A or B.
210. Matchmarking not as specified for Level A or B.
211. Matchmarking and information tags not as specified for Level A.
212. Unpainted exterior metal surfaces not coated with preservative for Level A or B.
213. Exposed gears not coated with preservative as specified for Level A or B.
214. Drive chains not preserved, and removed chains not coiled, wrapped, and secured as specified for Level A or B.
215. Exposed ball bearings not preserved as specified for Level A or B.
216. Pressure gages not packaged as specified for Level A or B.
217. Drive belts not removed or released from tension for Level A or B.
218. Unpainted pulley grooves not coated with primer as specified for Level A or B.
219. Electric motors not preserved in accordance with the referenced document for Level A or B.
220. Solution feeder assembly not preserved and tagged as specified for Level A or B.
221. Electric wiring not coiled and secured as specified for Level A or B.
222. Openings into electrical fixtures on the control panel not sealed with tape as specified for Level A or B.
223. Loose electrical fittings not preserved and packaged as specified for Level A or B.
224. Float switches, floats, and float rods not packaged as specified for Level A or B.
225. Threads of pipe and pipe fittings not protected with thread protectors as specified for Level A or B.
226. Sand and gravel not packaged in bags specified for Level A or B.
227. Net weight of bags of sand and gravel exceeds 100 lbs for Level A or B.
228. Repair parts not preserved in accordance with the referenced document for Level A or B.
229. Maintenance tools not preserved and packaged in accordance with the referenced document for Level A or B.
230. Consolidated packaging not as specified for Level A or B.
231. End sections of the erdlator not bolted together as specified for Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

232. Center section of the erdlator not reinforced as specified for Level A.
233. Blocking, bracing, and anchoring of the erdlator assemblies, sand filters, and other components not in accordance with the referenced document for Level A.
234. Packaged sand and gravel not palletized as specified for Level A.
235. Other components not packed as specified for Level A.
236. Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. Unless otherwise specified (see 6.2), the supplier shall furnish a preproduction pack for examination and test within the time frame specified to prove, prior to starting production packaging, that the applied preservation, packaging, packing, and marking comply with the preparation for delivery requirements of this specification (see 6.5). Examination and tests shall be those specified herein (see 4.8).

5.2 Preservation and packaging. Preservation and packaging shall be Level A, B, or C as specified.

5.2.1 Caution. Contact preservatives shall not be applied to any surfaces, other than the interior of the pump, which will come in contact with chemical solutions, slurries, or water, except waste water.

5.2.2 Level A.

5.2.2.1 Disassembly. The components and assemblies of the unassembled plant shall be disassembled in accordance with the recommendations of the plant manufacturer to the minimum extent necessary to accomplish protection against damage and loss and to attain minimum practical packing cube. Removed bolts, pins, nuts, and washers shall be replaced in one of the mating parts and secured to prevent their loss, or packaged together in a close-fitting box conforming to PPP-B-636, W5c, minimum.

5.2.2.2 Matchmarking. To facilitate reassembly, removed components and parts shall be matchmarked. Matchmarking information shall be on tags conforming to UU-T-81, Type A. The information shall be applied with a waterproof ink and the tags attached to the components so as to prevent damage to the preservation or to the components.

5.2.2.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

5.2.2.4 Unprotected surfaces. Unpainted exterior metal surfaces of components such as bolts, nuts, linkages, hose clamps, adapters, couplings, and surfaces of similar components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative.

5.2.2.5 Exposed gears. Exposed gears shall be coated with Type P-1 preservative or with primer conforming to TT-P-664, Composition G.

5.2.2.6 Drive chains. Drive chains shall be coated with enough Type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. The excess preservative shall be drained, and the entire chain shall be coated with Type P-1 preservative. Removed chains shall be coiled, and the coil shall be secured with soft, annealed wire and wrapped with barrier material conforming to MIL-B-121, Type II, Grade A, class optional, secured with tape conforming to PPP-T-60, Type IV. Sprockets and other unprotected surfaces of the drives shall be coated with Type P-1 preservative.

5.2.2.7 Exposed ball bearings. Exposed surfaces of ball bearings shall be coated with Type P-11 preservative. The bearings shall be wrapped or covered with barrier material conforming to MIL-B-121, Type II, Grade A, class optional, secured in place with tape conforming to PPP-T-60, Type IV.

5.2.2.8 Pressure gages. Pressure gages that are not installed, shall be packaged in a close-fitting fiberboard box conforming to PPP-B-636, W6c, minimum. Each gage shall be cushioned within the box to prevent movement and damage.

5.2.2.9 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved in accordance with MIL-P-116, Method IC-1. Unpainted pulley grooves shall be coated with primer conforming to TT-P-664.

5.2.2.10 Electrical components.

5.2.2.10.1 Electric motors. Electric motors shall be preserved in accordance with MIL-E-16298, Level A, by the alternate method as specified therein for assembled machines, except for the solution feeder assembly motor (see 5.2.2.10.2).

5.2.2.10.2 Chemical solution feeder assembly. The chemical solution feeder assembly gear-reduction mechanism and motor housing shall be filled to the operating level with oil conforming to MIL-L-21260, Type I, Grade 10. The gears shall be actuated to assure coating of all parts and surfaces within the housing. A tag conforming to UU-T-81, Type A,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

shall be attached in a conspicuous location indicating, "The gear housing is filled with the lubricant required for operation." The information on the tag shall be applied with waterproof ink.

5.2.2.10.3 Electric wiring. Electric wiring required for each plant shall be coiled to not less than 10 times its diameter. Each coil shall be secured with evenly spaced wraps of cotton tape or twine.

5.2.2.10.4 Electrical control panels. The openings into electrical fixtures on the control panels shall be sealed with tape conforming to PPP-T-60, Type IV.

5.2.2.10.5 Electrical fittings. Loose electrical fittings shall be preserved in accordance with MIL-P-116, Method IC-1, and all loose fittings packaged together in a close-fitting fiberboard box conforming to PPP-B-636, W6c, minimum, as applicable to the weight of contents.

5.2.2.11 Float switches. Float switches not installed on the tanks shall be packaged in a close-fitting fiberboard box conforming to PPP-B-636, W6c, minimum. Each switch shall be cushioned within the box to prevent movement and damage. The floats and float rods shall be consolidated packaged as specified in 5.2.2.18.

5.2.2.12 Pipe and pipe fittings. External threads of pipe and pipe fittings shall be protected with metal or plastic thread protectors.

5.2.2.13 Sand and gravel. The sand and gravel shall be packaged in bags conforming to PPP-B-35, Type III, Class 2 or 4, style optional, Bag Number P-15 or P-35. The net weight of each bag shall not exceed 100 pounds.

5.2.2.14 Repair parts. The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.2.15 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, Level A.

5.2.2.16 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1.

5.2.2.17 Other components. Other components not specifically mentioned herein shall be preserved as specified for similar components specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

5.2.2.18 Consolidated packaging. All component items and assemblies except tanks, erdlator sections, sand filters, sand and gravel and items specified in 5.2.2.19 shall be consolidated packaged in boxes conforming to PPP-B-601, Domestic Type, style optional, or PPP-B-621, Class 1, style optional. The pipe, pipe fittings, valves, and flanges designated by the color code shall be packaged together in one or more containers as applicable to the weight and size (see 5.2.2.19). The components shall be cushioned, blocked, or braced as applicable within the boxes in a manner to prevent movement and damage in accordance with MIL-STD-1186. Strapping shall be in accordance with the appendix to the applicable box specification, except boxes not exceeding 200 pounds gross weight shall not require strapping.

5.2.2.19 Consolidated packaging exceptions. To the extent that space permits, unboxed pipe, pipe fittings, valves and flanges shall be packed with the erdlator sections, secured to the crate base. Use of dunnage to separate noncompatible components is recommended.

Note: A particularly large space is available at the front and rear ends of the crate for the erdlator sections (see 5.3.1.1).

5.2.3 Level B (see 6.6). The components comprising a complete plant shall be preserved and packaged as specified for Level A with the following exceptions.

5.2.3.1 Fiberboard boxes. Fiberboard boxes specified in 5.2.2.1, 5.2.2.8, 5.2.2.10.5 and 5.2.2.11 shall conform to PPP-B-636, Type CF, Class Domestic, Variety SW, grade as applicable.

5.2.3.2 Drive chains. Drive chains shall be preserved as specified for Level A, except only Type P-3 or P-9 preservative shall be applied.

5.2.3.3 Limestone and chemical solution feeder assemblies. The limestone and chemical solution feeder assemblies gear reduction mechanism and motor housing shall be filled to the operating level with the lubricant required for operation and the gears actuated to assure coating all parts and surfaces within the housing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

5.2.4 Level C. The components comprising a complete plant shall be preserved and packaged in accordance with 5.2.1 in a manner to afford protection against deterioration and damage from the suppliers to the initial destination. The supplier standard practice is acceptable when it fulfills these requirements.

5.3 Packing. Packing shall be Level A or C as specified.

5.3.1 Level A.

5.3.1.1 Erdlator assembly. The erdlator assembly, disassembled in sections, shall be packed in a crate, Type I, Class 2, Style A, of MIL-C-104. The two end sections of the erdlator shall be bolted together through the flanges of the sections. The two launder ring sections shall be secured to and shipped with the two erdlator sections. The center section shall be reinforced on each side with a steel stiffener plate or several steel flat bars bolted to the section at opposite points to prevent distortion of the section during handling and shipment. Blocking, bracing, and anchoring of the erdlator assemblies shall be in accordance with the appendix to MIL-C-104 and with MIL-STD-1186.

5.3.1.2 Sand filters, chemical solution tanks, limestone slurry feeder, and erdlator agitator assembly. The two sand filters shall be inverted and packed together in a Type I, Class 2, Style A MIL-C-104 crate. Dunnage shall be used under the perimeter of the tank bases to compensate for the angle iron brace across the maximum diameter of the tanks. Tanks shall be secured to the base by clamps or by bolts through the perimeter angle iron. Prior to mounting the tanks on the skid base but after their location has been accurately determined, the two chemical solution tanks shall be inverted and positioned one each under each filter tank so positioned to be at the outside half section of each tank, i.e., at the extreme ends of the crate. The limestone slurry feeder and the erdlator agitator assembly shall be located under the innermost half section of each tank so that they will be in the center of the crate. All items located under the filter tanks shall be bolted and secured to the base of the crate to prevent movement during shipping and handling in accordance with requirements of MIL-STD-1186.

5.3.1.3 Sand and gravel. The packaged sand and gravel shall be packed in accordance with MIL-STD-147, Load Type XV, on pallets conforming to NN-P-71, Type IV, the top of which will be reinforced with 1/4-inch plywood conforming to NN-P-530.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

5.3.1.4 Other components. All other components, consolidated packaged and unpackaged, shall be packed in minimum number of nailed wood crates conforming to MIL-C-104, Type I, Class 2, Style A. Blocking, bracing, and anchoring of the contents shall be in accordance with the appendix to the crate specification and with MIL-STD-1186.

5.3.2 Level C. The components of the plant shall be packed to insure carrier acceptance and safe delivery to destination at lowest rating in containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules. Each side of the center section of the erdlator shall be reinforced on each side at opposite points as specified in 5.3.1.1 or in any other suitable manner.

5.4 Marking. Packages, shipping containers, skid-mounted components, and palletized loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The military design plant covered by this specification is intended for use with ancillary items referenced in appendix to supply 10,000 gph of potable water from a raw, fresh water source.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specfication.
(b) Time frame required for submission of first article (preproduction model) (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(d) When initial production inspection is required, and when the plant is to be furnished when applicable (see 3.3).
(e) **When the Government will furnish ancillary items specified in the appendix** (see 3.32).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

(f) Time frame required for submission of the preproduction pack (see 5.1).
(g) When preproduction pack is not required (see 5.1).
(h) Level of preservation and packaging and level of packing required (see 5.2 and 5.3). Level B preservation and packaging is intended to provide economical but limited protection and should be specified only when it is determined the plants will be held in covered storage.

6.3 Preproduction model. Any changes or deviations of production plants from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish plants conforming to this specification (see 3.3).

6.4 Health hazard determinations. For health determinations (see 3.4), advice should be requested from the following:

(a) Army - The Surgeon General, (MD).
(b) Navy - Chief, Bureau of Medicine and Surgery, (MS).
(c) Air Force - The Surgeon General, (03).

6.5 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the plants in accordance with this specification (see 5.1).

6.6 Level B packaging has been developed by the Army and Air Force for specific conditions specified in 6.2. For Navy procurements, details of Level B preservation and packaging, when required, should be developed at the time of procurement when all conditions are known (see 5.2.3).

6.7 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and supplier's maintenance and operation manual to be furnished with each plant.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of plants.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:07Z
Check the source to verify that this is the current version before use.

6.9 Lubricants. The contracting officer should furnish a list of military lubricants (see 3.26.1) applicable to the plant covered by this specification as contained in the Federal Supply Catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.10 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.32.

6.11 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections.

6.11.1 Definition of SQAP. A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

Custodians:

Army - ME
Air Force - 84

Review activities:

Army - CE
Navy - MS
Air Force - 03
DSA - CS

User activities:

Army - CE
Navy - MC, YD

Preparing activity:

Army - ME

Project No. 4610-0072

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

APPENDIX

10 Scope. This appendix provides for information of a typical set of ancillary items required to support the function of the plant.

Note: The consumable items listed in 30.1 provide for a maximum of 30 days operation (600 hours). For periods of operation beyond 30 days, quantitative adjustments of the consumable items should be made.

20 Applicable documents.

20.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this appendix to the extent specified herein:

SPECIFICATIONS

Federal

L-H-520	- Hose and Hose Assemblies, Nonmetallic (Rubber, Plastic).
O-C-114	- Calcium Hypochlorite, Technical and Chlorinated Lime, Technical.
W-L-661	- Light, Extension, Electrical.
HH-P-151	- Packing, Rubber-sheet, Cloth-Insert.
RR-S-131	- Scoop, Kitchen, Corrosion-resisting Steel, 7-7/8 Inches Long, 5-1/4 Inches Wide.
WW-C-624	- Coupling Assembly Hose (Garden Water and Water Suction).
WW-V-58	- Valves, Gate, Cast iron, Threaded, and Flanged (For Land Use).
ZZ-H-561	- Hose, Rubber, and Hose Assemblies, Rubber Smooth Bore, Water Suction and Discharge.
ZZ-H-601	- Hose and Hose Assemblies, Rubber (Yarn and Fabric Reinforced) Water Service.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

Military

MIL-C-506	- Carbon, Activated, Powdered.
MIL-C-538	- Comparator, Color.
MIL-C-3432	- Cable and Wire, Electrical (Power and Control, Semi-Flexible, Flexible and Extra Flexible, 300 and 600 Volts).
MIL-T-10086	- Tanks, Liquid Storage, Metal; Vertical, Bolted.
MIL-C-10387	- Coupling, Clamp, Pipe; with Bolts and Synthetic-Rubber Gasket, for Grooved Pipe and Tube.
MIL-P-10388	- Pipe Fittings: One or More Ends Grooved.
MIL-P-11087	- Pipe, Steel: Grooved, Threaded, or Plain Ends.
MIL-S-12165	- Strainers, Suction, Fire-Hose; and Strainers, Suction, Hose.
MIL-H-12732	- Hypochlorination Unit, Water Purification, Frame Mounted, Automatically Controlled 2 to 100 GPM Flow.
MIL-P-14514	- Pump, Centrifugal, Self-priming, Frame-mounted, 1-1/2-INCH, 65-GPM at 50 Feet Head (For Water).
MIL-L-14519	- Limestone, Pulverized.
MIL-F-14580	- Ferric Chloride, Anhydrous, Crystalline, Technical.
MIL-H-52036	- Heater, Water, Oil-fired.
MIL-P-52415	- Pail, Rubber and Plastic, Graduated.

STANDARDS

MS 51967	- Nut, Plain, Hexagon-carbon Steel, Cadmium Plated, UNC-2B (In./MM).
MS 90725	- Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNC-2A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

DRAWINGS

ME

D13200E9060	- Typical Plant Layout & Functional Components.
PL-13200E9060	- Water Purification Plant Functional Components.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

20.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC.

B 16.1	- Cast Iron Pipe Flanges and Flanged Fittings.

30 Functional components of 10,000 GPH water purification plant.

30.1 Components for 30 days (600 hours) operation.

Item	Identifying No.	Description	U/I	Qty
(1)	NSN	- Cable, Power, Electrical: ptbl cord; 600 v; 4 copper conductors; No. 4 AWG; stranded; 133 strands; No. 25 AWG; w/gnd wires; synthetic rubber insulated; covering in sequence from conductors is synthetic rubber; 81349----CO-04 HLF (3/2-3/10R)1340-C-3432.	Ft	100
(2)	6810-255-0472	- Calcium Hypochlorite, Technical; 100 lb, Moisture-proof drum; 81348----O-C-114, Type 1.	Dr	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

(3)	6810-264-6575	- Carbon, Activated, Powdered; 50 lb, moisture-proof drum; 81349----MIL-C-506.	Dr	2
(4)	6630-087-1838	- Comparator, Color; 81349----MIL-C-538.	Ea	1
(5)	6810-682-6677	- Ferric Chloride, Anhydrous, Crystalline, Technical; 100 lb, moisture-proof drum; 81349----MIL-F-14580.	Dr	18
(6)	6115-118-1247	- Generator Set, Diesel Engine: 30 MEP-Mdl No. 104A, 120/208, 240/416 v, 3 phase, 50/60 Hertz, make and model unspecified.	Ea	1
(7)	6810-087-2340	- Indicator Solution, pH Wide Range: 16 oz plastic bottle; 79172----UXE18895.	Bt	1
(8)	6230-162-1227	- Light, Extension; 2 conductor; Type SO or STO, 16 AWG cable, 50 ft lg, std 2 parallel blade, plug connector one end; lampholder, guard, hook, reflector, rubber handle w/switch other end; 100 w medium screw base lamp accommodated; 81348----W-L-661, Type III.	Ea	1
(9)	NSN	- Light, Extension; 2 conductor; Type SO or STO, 14 AWG cable w/grdn wire, 50 ft lg, std 3 parallel blade plug connector one end, other end 100 w spot beam screw base lamp accommodated. (Commercial); N/A	Ea	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

(10)	6810-585-2017	- Limestone, Pulverized; calcium carbonate, 94 pct min, powder; USS sieve No. 100, 4 pct max retained, No. 325, 20 pct max passing; 80 or 100 lb bag; 81349----MIL-L-14519.	Bg	12
(11)	6810-270-8289	- O-Tolidine Dihydrochloride Solution; 16 oz plastic bottle; 79172----UXA7787.	Bt	1
(12)	7240-246-1097	- Pail, Utility: 3 gal; steel handle, plastic or rubber covered; w/pouring lip; 81349----MIL-P-52415, Size 2, Grade B.	Ea	1
(13)	7330-153-9760	- Scoop, Kitchen; cres; 5.25 in w, 7.87 in lg; 81348----RR-S-131.	Ea	2
(14)	6810-986-3471	- Sodium Arsenite Solution; 16 oz plastic bottle; 79172----UXA9931.	Bt	1

30.2 Components required for extreme cold climate operation.

(1)	4520-289-7976	- Heater, Water, Oil; automatically controlled; high pressure atomizing burner; No. 2 fuel oil; 3,000 gal of water heated per hr, 40 deg temp rise per hr; one 1 in and two 1-1/2 in - 11-1/2 NPT outlets; 150 psi wp; 74 in lg X 38 in w X 140-7/8 in h, w/stack; 208 v, 3 phase, 50/60 Hz; 81349----MIL-H-52036.	Ea	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

(2)	4720-202-6722	- Hose, Assembly, Nonmetallic: Water; brass couplings each end; rigid external thd one end, swivel internal thd other end; braided; 150 psi test pressure; hex nut; barbed insert couplings, 3/4 in - 14 NPSH, WW-C-624, Type B; 3/4 in id; 50 ft lg, excl couplings; 81348----L-H-520, Type I.	Ea	2
(3)	4720-202-6731	- Hose Assembly, Nonmetallic: Wrapped; wire reinforced; 300 psi tested pressure; 1-1/2 in-11-1/2 NPSH; 1-1/2 in id; 10 ft lg, excl couplings; 81348----ZZ-H-561, Type I, Grade B.	Ea	10
(4)	4320-289-7977	- Pump, Centrifugal: fresh water; elec mtr driven, ac, 208 v, 3 phase, 50/60 Hz, 2 hp; frame mtd; 1-1/2 in inlet, 1-1/2 in outlet, thd; 65 gpm at 50 ft head; make and model unspecified; 81349----MIL-P-14514, Type I.	Ea	1
(5)	4730-684-4296	- Strainer, Suction, Hose: brass, bbl; 1-1/2 in - 11-1/2 NPSH, internal; 4-1/2 in lg; hex or octagon wrenching feature; No. 35 inside cut V-slots; 81349----MIL-S-12165, Type II.	Ea	1

30.3 Components recommended for complete water purification plant assembly.

(1)	4610-269-0165	- By-pass, Connection Assembly; Hypochlorination Unit; 81349----MIL-H-12732.	Ea	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

(2)	4730-142-1590	- Coupling, Pipe, Grooved Type, 4 in, grooved both ends; w/bolts, nuts, and gasket, for coupling clamp; 81349----MIL-C-10387.	Ea	60
(3)	4730-273-8299	- Elbow, Pipe, Grooved Type; MI, 90° X 4 in , size; 81349----MIL-P-10388.	Ea	15
(4)	NSN	- Flange; CI, 4 in X 10-1/2 in, size; ANS-B16.1.	Ea	4
(5)	4370-174-5921-CI or 4370-221-6253-MI	- Flange, Companion, Pipe; 4 in, CI or MI: - ANS-B16.1.	Ea	13
(6)	NSN	- Gasket, Full Face; Cloth Inserted Rubber, 4 in id X 9 in od X 1/16 in thk; 81348----HH-P-151	Ea	13
(7)	NSN	- Gasket, Full Face; Cloth Inserted Rubber, 4 in id X 10-1/2 in X 1/16 in thk; 81348----HH-P-151.	Ea	4
(8)	4610-269-0163	- Hypochlorination Unit, Water Purification, Frame Mtd, Automatically Controlled 2 to 100 GPM flow; 81349----MIL-H-12732.	Ea	1
(9)	5310-763-9820	- Nut, Plain, Hex; Cadmium Plated, Stl; 5/8-11UNC-2B; 81349----MS 51967-20.	Ea	104
(10)	4730-202-7212	- Nipple, Pipe Fittings, Adapter, Grooved one end, other end threaded; 4 in-8 NPT, Stl; 81349----MIL-P-10388.	Ea	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

(11)	4320-376-6957	- Pump, Centrifugal, Self-Priming; 500 GPM, 4 inch, GED; 10912----Pt. No. 30MCCG.	Ea	1
(12)	NSN	- Pipe, Steel, Grooved, Both Ends; 4 in X 48 in lg, Stl; 81349----MIL-P-11087.	Ea	4
(13)	NSN	- Pipe, Steel, Grooved, Both Ends; 4 in X 8 ft - 3 in lg, Stl; 81349----MIL-P-11087.	Ea	2
(14)	NSN	- Pipe, Steel, Grooved, Both Ends; 4 in X 36 in lg, Stl; 81349----MIL-P-11087.	Ea	2
(15)	NSN	- Pipe, Steel, Grooved, Both Ends; 4 in X 16-1/2 in lg, Stl; 81349----MIL-P-11087.	Ea	2
(16)	NSN	- Pipe, Steel, Grooved, One End, Other End Threaded; 4 in X 18 in lg, Stl; 81349----MIL-P-11087.	Ea	2
(17)	NSN	- Pipe, Steel, Grooved, One End, Other End Threaded; 4 in X 9 in lg, Stl; 81349----MIL-P-11087.	Ea	2
(18)	4710-202-7906	- Pipe, Steel, Grooved, One End; Cut to suit; 81349----MIL-P-11087	Ft	200
(19)	DWG-11005-1 thru 5 (Reference)	- Platform and Foundation Layout and Equipment Elevations; 97403----ATTN: AMXFB-DPS	Sht	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

(20)	5305-724-5914	- Screw, Cap, Hexagon-carbon steel, cadmium plated; 5/8-11UNC-2A X 3 in lg; 81349----MS 90725-168.	Ea	104
(21)	NSN	- Tee, Pipe, Grooved Ends; 4 in, MI; 79154----Pt. No. 20.	Ea	3
(22)	4820-289-3379	- Strainer, Suction, Pipe; 4 in - 8 NPT, Internal Thread; N/A	Ea	1
(23)	DWG-13200E9060 (Reference)	- Typical Plant Layout and Functional Components; 97403----ATTN: AMXFB-DPS	Sht	1
	PL-13200E9060	- Water Purification Plant Functional Components	Un	1
(24)	5430-263-6077	- Tank, Liquid Storage, Metal, Vertical, Bolted; 500 bbl (21,000 gal). 81349----MIL-T-10086, Type I	Ea	1
(25)	5430-263-6080	- Tank, Liquid Storage, Metal, Vertical, Bolted; 250 bbl (10,500 gal). 81349----MIL-T-10086, Type I	Ea	2
(26)	4820-268-8773	- Valve, Cast-iron, Gate, 4 in 125 and 250, pound, Screwed and Flanged (For Land Use); 81348----WW-V-58.	Ea	4
(27)	NSN	- Warehouse, Prefabricated, Building, System, 40 ft X 40 ft; or equivalent, (Reference). N/A	Ea	1

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:07Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.